

# VYBE TUTOR

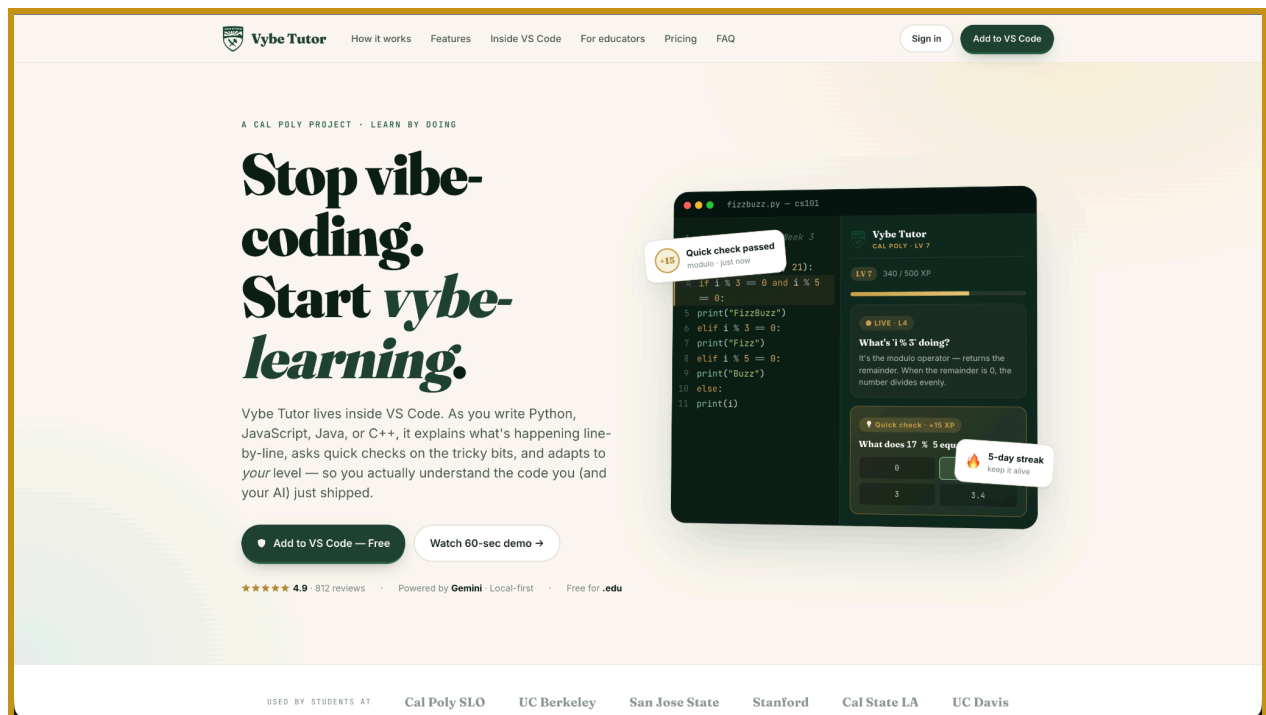
## Product Requirements Documentation

**Target Release:** May 3rd, 2026 | **Status:** In Active Development | **Platform:** VS Code Extension  
**Developers:** Naomi, Brian, Tan, Valerie

## Goals

Vybe Tutor aims to:

- Reduce the gap between **AI-assisted code generation ("vibe coding")** and **genuine conceptual understanding**.
- Enable students to actively learn while coding by providing **real-time**, contextual explanations directly in your **IDE's sidebar**.
- **Increase engagement** and retention through **gamified elements** such as quizzes, XP points, levels, and daily streaks.
- **Integrate seamlessly into your IDE** so the learning layer adds value **without disrupting** existing coding workflows.
- Personalize explanations and content based on **user skill level**, **quiz performance**, and **imported course materials (e.g., syllabi, lecture notes)**.



## Background

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AI-assisted coding tools have dramatically accelerated development workflows, especially for students and early-career developers. However, this speed comes at a cost: many users accept and ship generated code without understanding how or why it works. This creates a growing gap between code production and conceptual mastery, which is particularly prevalent among computer science students.

Vybe Tutor is designed to close this gap by transforming passive "vibe coding" into an active learning experience. Rather than replacing the coding effort, Vybe Tutor augments it by embedding a real-time teaching layer directly inside your preferred IDE. Through a sidebar extension, the tool delivers live explanations, prompts comprehension checks, and reinforces knowledge through gamified mini-games and quizzes.

Unlike traditional learning platforms (flashcards, tutorials, or external courses), Vybe Tutor operates in-context alongside the user's actual code. This ensures that learning is immediately relevant, applied, and retained. By supporting custom inputs like lecture notes and syllabi, the system tailors explanations to align with a student's academic environment.

## Assumptions

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- Users are already working in an IDE (e.g., VS Code) and are willing to install and use extensions within their existing workflow.
- There is growing demand for AI-assisted tools that prioritize conceptual learning alongside productivity.
- Users benefit more from in-context, real-time explanations than from separate external learning platforms.
- Gamification elements (XP, levels, streaks, quizzes) will increase engagement and motivation without meaningfully disrupting coding flow.
- Users are willing to import external learning materials (syllabi, lecture notes) to improve personalization.
- The AI system can generate accurate, appropriately leveled explanations across multiple programming languages (Python, JavaScript, Java, C++).
- API response latency will be low enough to deliver explanations within 2-3 seconds of a code event, avoiding workflow disruption.

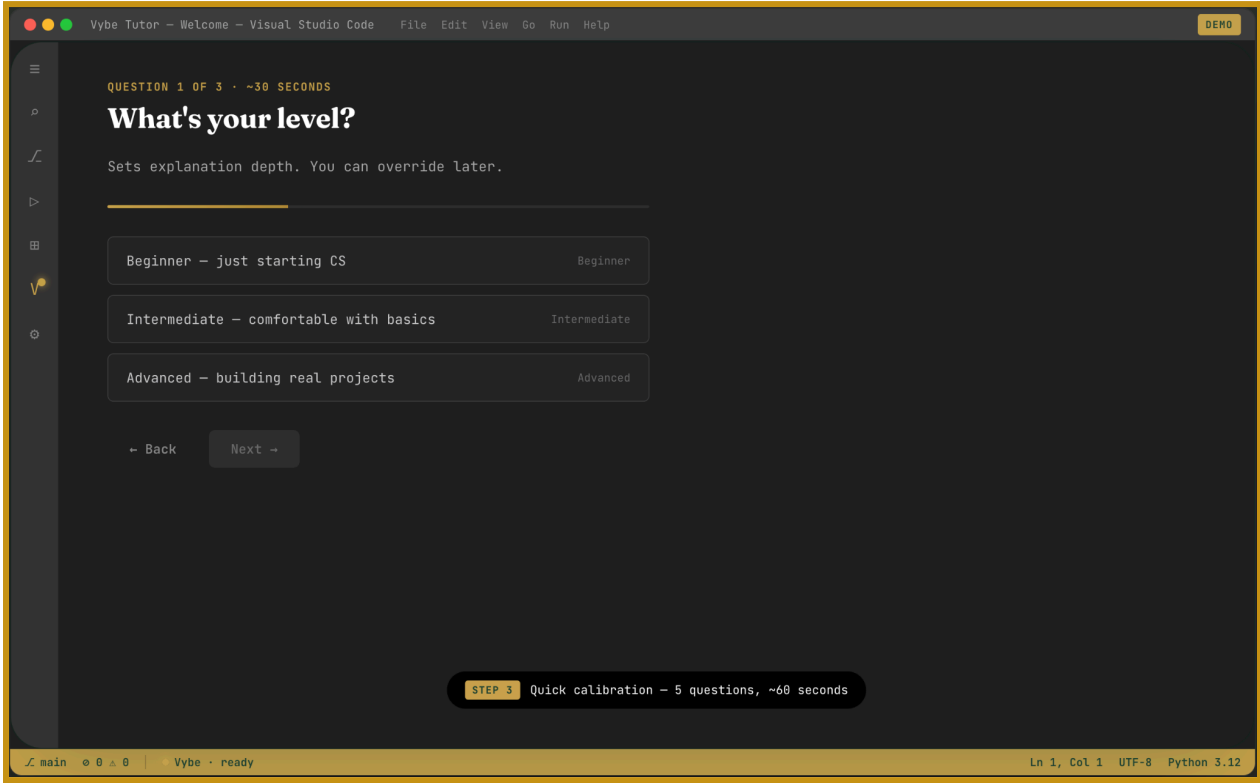
## User Types

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Vybe Tutor is designed for students and early-career developers across skill levels. The primary user segments are:

| User Type                                 | Description                                                                                                                                                | Primary Need                                                                                                                       |
|-------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| <b>Beginner<br/>(Just Starting in CS)</b> | New programmers who are learning syntax, logic, and foundational concepts for the first time. They often follow tutorials or rely on AI-generated code but | Step-by-step, simplified explanations that connect code to core programming concepts and build confidence through guided learning. |

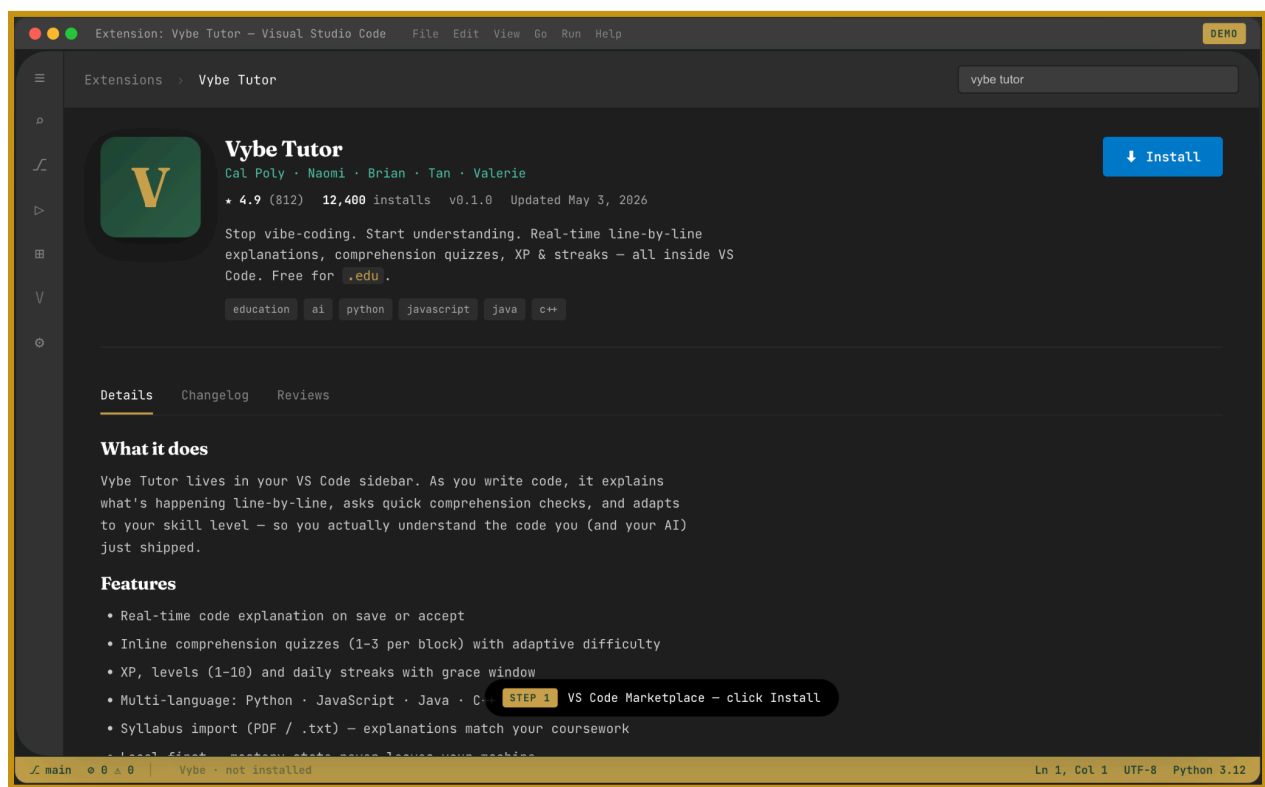
| User Type                                     | Description                                                                                                                                                                                                            | Primary Need                                                                                                                                                             |
|-----------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                               | struggle to connect code to underlying concepts.                                                                                                                                                                       |                                                                                                                                                                          |
| <b>Intermediate<br/>(Comfortable with CS)</b> | Students who can write functional code and complete assignments but are transitioning from guided learning to independent problem-solving. They are starting to encounter larger codebases and more abstract concepts. | Context-aware insights that explain <i>why</i> certain approaches are used, helping them bridge the gap between writing code and designing solutions.                    |
| <b>Advanced<br/>(Building Real Projects)</b>  | Developers working on complex or real-world projects who are focused on efficiency, scalability, and best practices. They are less concerned with syntax and more with architecture, patterns, and optimization.       | On-demand, high-level explanations and feedback that support better design decisions, code quality, and deeper technical understanding without slowing down development. |



# Use Cases

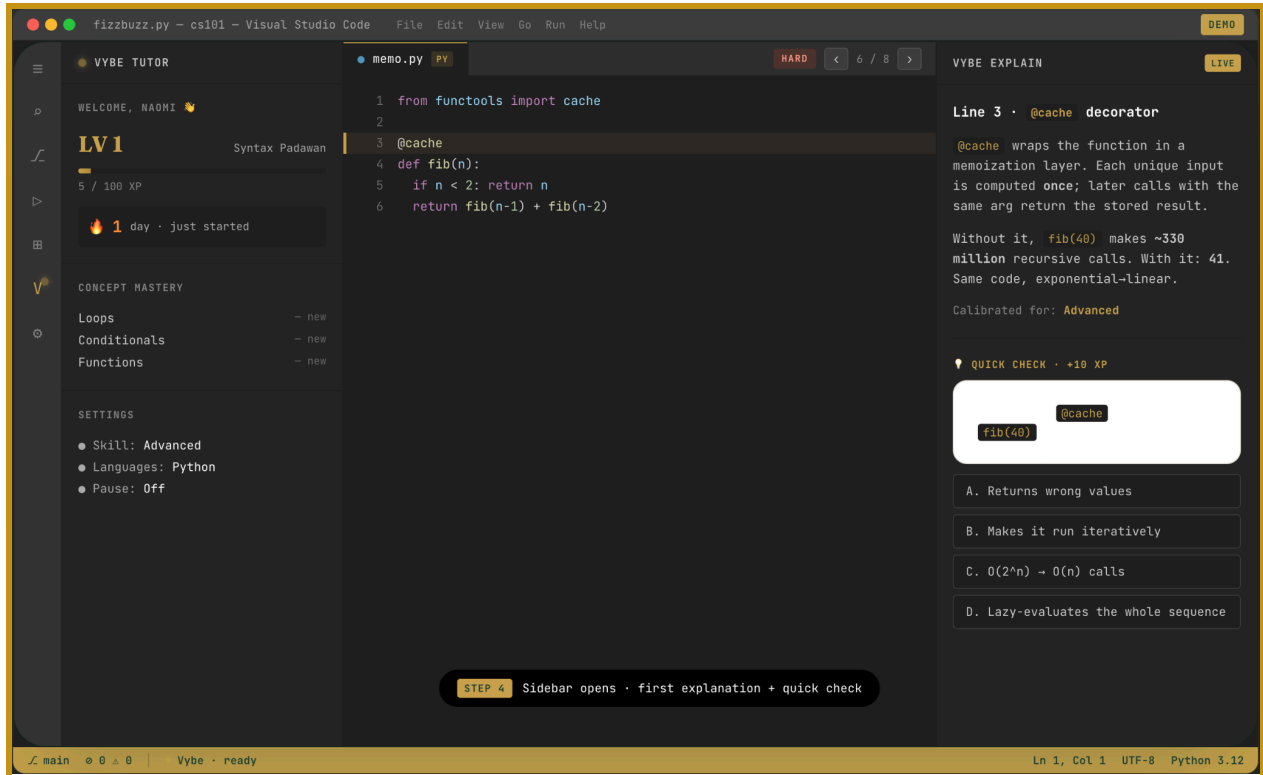
## Flow 1: Installation & Onboarding

1. Users discover Vybe Tutor via the Extension Manager of a VS Code.
2. The user clicks the "Add Extension" button on the extension page or the Vybe Tutor landing page.
3. VS Code prompts confirmation; user installs the extension.
4. On first activation, the user is shown a brief skill-level onboarding quiz (3-5 questions) to calibrate explanation depth.
5. Sidebar panel opens; user writes or pastes a code snippet to trigger their first live explanation.
6. A comprehension check question appears; the user answers and earns their first XP.



## Flow 2: Active Coding Session

1. User opens an existing project in VS Code.
2. The user writes new code or accepts an AI code suggestion.
3. On save, Vybe Tutor's sidebar automatically generates a plain-language explanation of the changed code block.
4. The user reads the explanation; a comprehension check appears at the bottom of the sidebar.
5. User answers correctly → earns XP and sees streak counter increment. Incorrect answer → follow-up hint + retry.
6. The user continues coding; streak and level progress are visible at the top of the sidebar throughout.



## Flow 3: Pausing Session

1. The user is in a time-pressured coding session and does not want interruptions.
2. The user clicks the pause/snooze icon in the Vybe Tutor sidebar header.
3. Explanations and quizzes are silenced for a configurable duration (15 min, 1 hr, or until manually re-enabled).
4. Streak protection: if the user has already answered at least one quiz that calendar day, their streak is preserved during the pause window.

## Flow 4: Deep Dive Exploration

1. The user hovers over a specific line or block of code within the IDE.
2. Vybe Tutor displays a concise, plain-language explanation inline.
3. If the user needs more detail, they click the **“Ask more in-depth”** button in-line explanation box.
3. An input field appears in the sidebar; the user types a follow-up question about the code or concept.
4. Vybe Tutor generates a more detailed, context-aware explanation, including examples or breakdowns if applicable.
5. The user reviews the response and can optionally ask additional follow-up questions to further clarify their understanding.
6. Vybe Tutor takes note of the code/concepts from user input to reevaluate future quizzes.

# Requirements

Requirements are prioritized using the MoSCoW method.

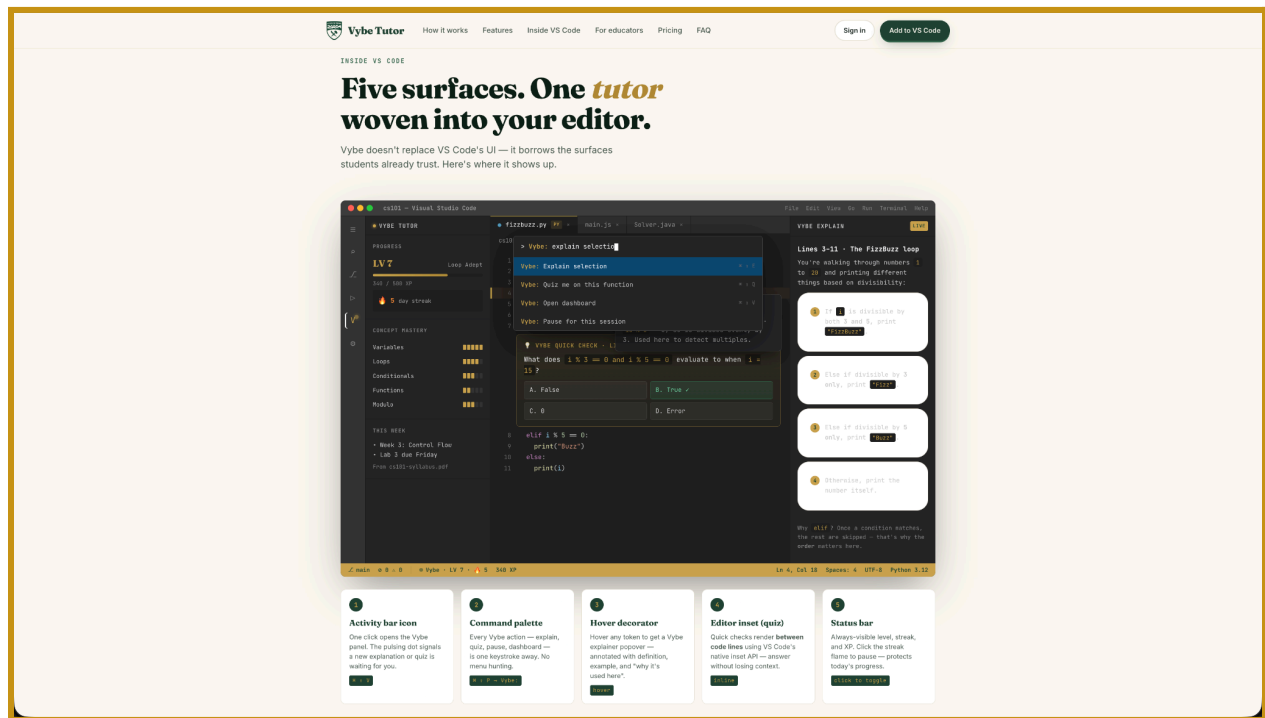
Color Coding: Red = Must Have, Yellow = Should Have, Green = Could Have.

| # | Title                                  | User Story                                                                                                                                  | Importance  | Notes                                                                        |
|---|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|-------------|------------------------------------------------------------------------------|
| 1 | Real-Time Code Explanation             | As a student, I want the sidebar to show an explanation of the code I just wrote or accepted so I understand what each part does.           | Must Have   | Triggered on code save/accept; uses Gemini API with code + context           |
| 2 | Comprehension Quiz / Check             | As a student, I want to be prompted with a short quiz question after a code explanation so I can verify my understanding before moving on.  | Must Have   | Multiple choice or fill-in; 1-3 questions per explanation block              |
| 3 | Gamification: Points, Levels & Streaks | As a student, I want to earn XP and maintain a coding streak so I feel motivated to keep learning while I build.                            | Must Have   | Daily streak resets at midnight; level badges displayed in sidebar           |
| 4 | Personalized Difficulty / Skill Level  | As a student, I want Vybe Tutor to adapt explanation depth to my skill level so I get beginner-friendly or advanced context as appropriate. | Should Have | Onboarding quiz sets baseline; adjusts over time via quiz performance        |
| 5 | Multi-Language Support                 | As a student, I want Vybe Tutor to explain code in Python, JavaScript, Java, and C++ so I can use it across all my courses.                 | Should Have | Language auto-detected from file extension; prompt includes language context |
| 6 | In-App Onboarding Walkthrough          | As a first-time user, I want a guided walkthrough when I install the extension so I know how to use every feature.                          | Could Have  | Step-by-step tooltips; skippable; re-accessible from settings                |

# User Interaction & Design

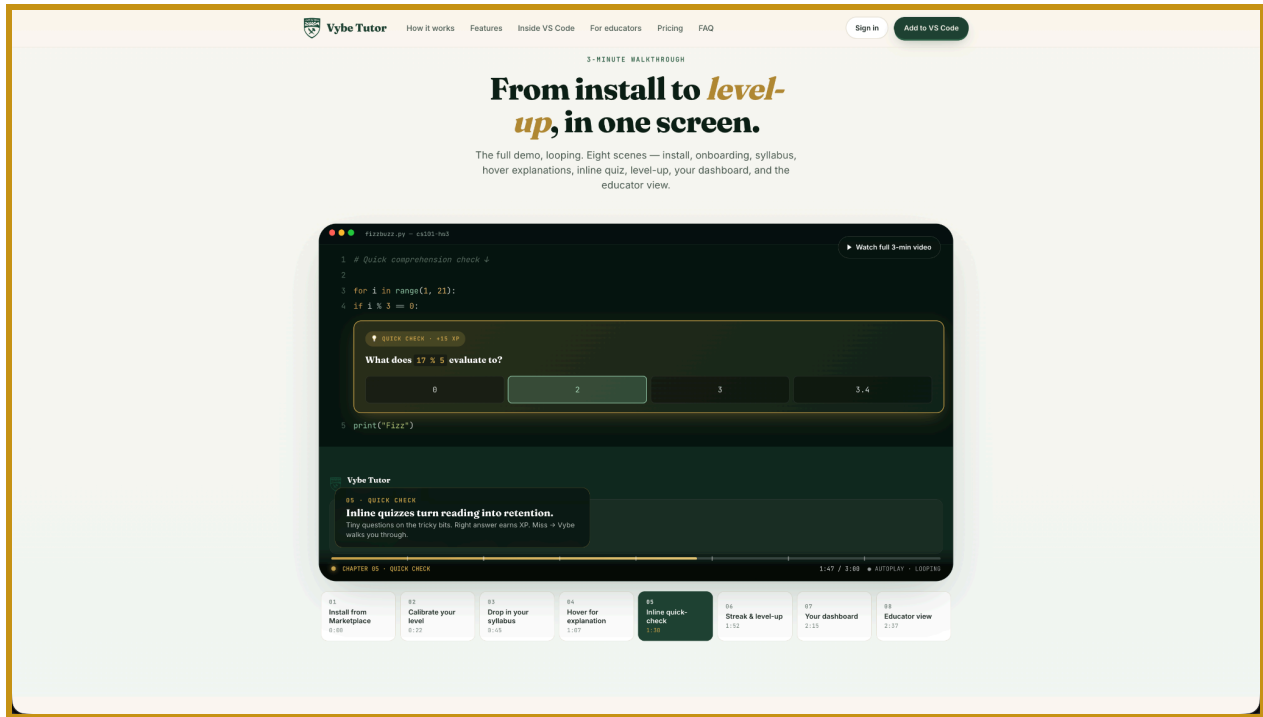
## Sidebar Layout

- Top bar: Vybe Tutor logo, current level badge, XP progress bar, streak counter with flame icon.
- Main panel: Scrollable explanation card for the most recently detected code block.
- Quiz section: Appears below the explanation card; multiple choice or short text input.
- Footer: Settings gear, pause/snooze toggle, import materials button.



## Gamification Design

- XP is awarded for correct quiz answers (10 XP), completing explanations (5 XP), and daily streaks (bonus multiplier).
- Levels progress from 1 (Green) through 10 (Rainbow); each level unlocks a color grading
- Streak resets at midnight local time; a 24-hour grace window is under consideration (see Questions).



## Explanation Card Design

- Explanation rendered in plain language with syntax-highlighted inline code references.
- "Why does this work?" toggle expands a deeper technical rationale for intermediate/advanced users.
- "Relate to my coursework" button (if materials imported) links the explanation to a relevant lecture topic.

## Release Criteria

### Functionality

- All Must Have requirements (items #1-4) must be implemented, tested, and passed QA.
- Extension is published and installable from the VS Code Marketplace.
- AI explanations are generated correctly for Python, JavaScript, Java, and C++ files.

### Usability

- Sidebar layout and visual design must match agreed mockups.
- An onboarding quiz must be completed in under 2 minutes for a new user.
- Explanation latency must be under 3 seconds on a standard broadband connection.

### Reliability & Performance

- Extension must not increase VS Code memory usage by more than 80 MB at idle.

- API error rate must be below 1% during peak usage; graceful fallback on failure.
- Streak and XP state must persist across VS Code restarts.

## Questions

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| Question                                                                              | Proposed Outcome / Resolution                                            |
|---------------------------------------------------------------------------------------|--------------------------------------------------------------------------|
| How should explanation depth be calibrated when no skill level data is available yet? | Default to beginner-friendly; update after first quiz attempt            |
| What happens to streaks when a user doesn't code for a day — grace period?            | TBD — consider a 24-hour grace window or streak freeze item              |
| Does Canvas import require institutional login, or is file upload sufficient for v1?  | File upload (.pdf, .txt) for v1; Canvas OAuth in future sprint           |
| How do we handle code explanations for files > 500 lines — chunk or truncate?         | Send relevant code block (function/class scope) + 10-line context window |
| What is the fallback when the API call for an explanation fails mid-session?          | Show cached last explanation + retry button; log error silently          |

## Future Implementations (v1 Scope Exclusions)

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- Canvas LMS OAuth integration.
- File Import (PDF, .docx, .txt) — course-material integration.
- Extension Install CTA — direct VS Code Marketplace “Add Extension” download button.
- Mobile or web-based companion app.
- Multi-user / classroom instructor dashboard.
- Real-time collaborative coding explanations.
- Support for IDEs other than VS Code (JetBrains, Cursor, etc.) in this release.